A Finite-Rate Selection Law for Annealed Gibbs Sampling

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Paper · code · interactive edition
github.com/ali-izhar/finite-rate-selection-law

PLAUSIBLE SAMPLES, WRONG PROPORTIONS

In Curie–Weiss and Hopfield models, annealed Gibbs samples keep only a fraction c of the model's log-odds between modes.



Fig. 1a · 14 paths of the drift-plus-noise approximation · N = 1,024 · G = 0

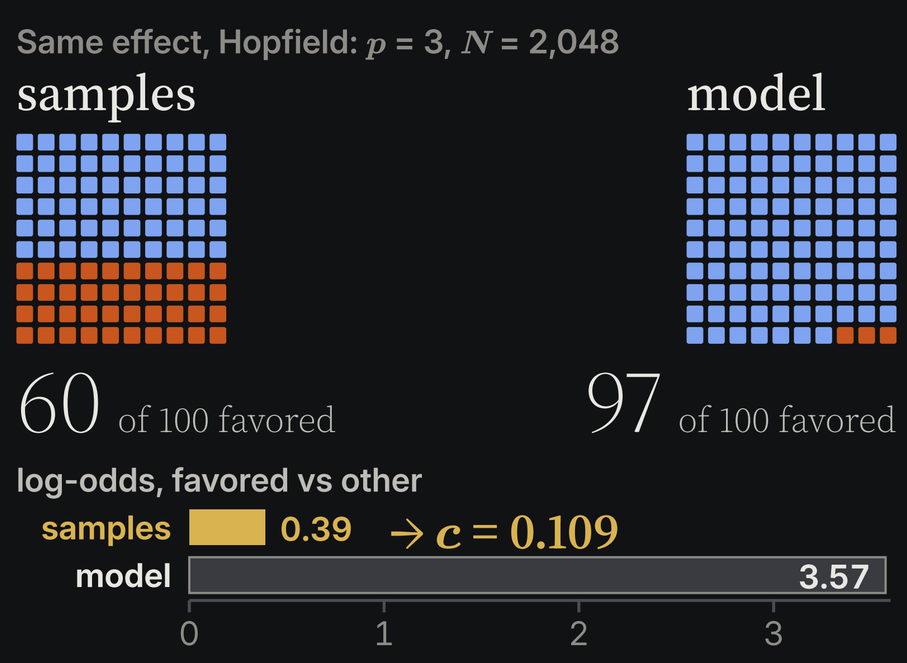
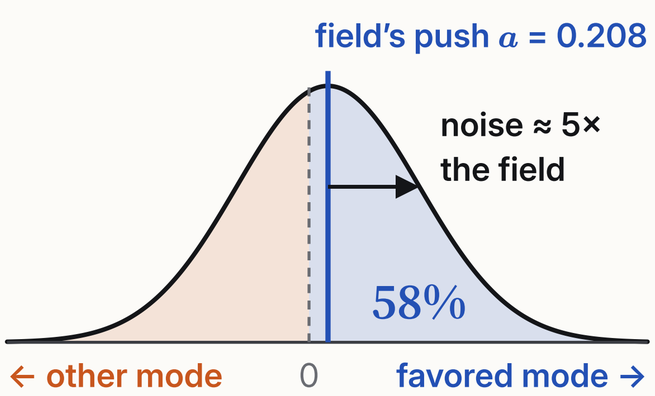


Table 1 · three patterns · N = 2,048 · cooling rate γ = 0.004

I Curie–Weiss: the exact chain tests the law

1 A weak field races noise.

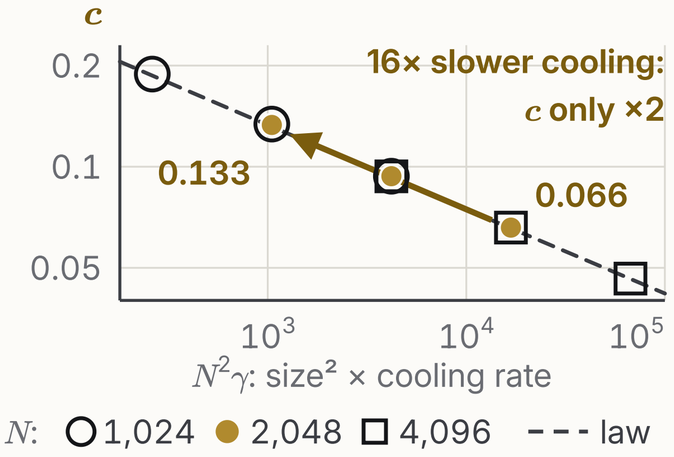


$$q_{\mathrm{sel}} \simeq \Phi(a), \quad a = G\sqrt{N}\,(\pi/\gamma)^{1/4}$$

classical result (Kondepudi & Nelson 1983)

q: share of runs in favored mode · G: field
Fig. 1b · N = 2,048 · cooling rate γ = 0.001

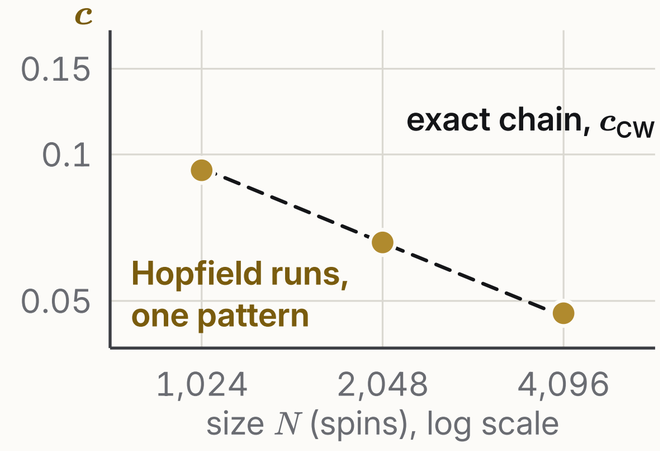
2 Slower cooling helps little.



$$c = \frac{\operatorname{logit} q_{\mathrm{chain}}}{\operatorname{logit} q_{\mathrm{eq}}} \propto (N^2\gamma)^{-1/4}$$

chain: sampler · eq: model · Table 2
linear ramps, NG fixed · law within 2%

3 One pattern is Curie–Weiss.

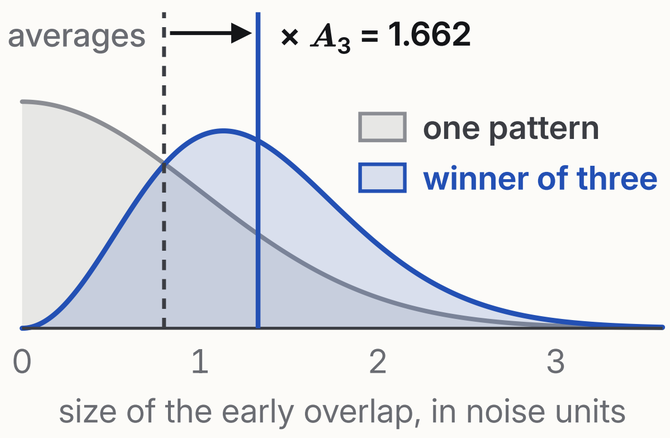


Hopfield runs with one pattern reproduce the exact chain.

Table 1 · p = 1 · ratios 0.990–1.015 · γ = 0.004

II Hopfield: competing patterns, predicted without Hopfield samples

4 Competition raises c.

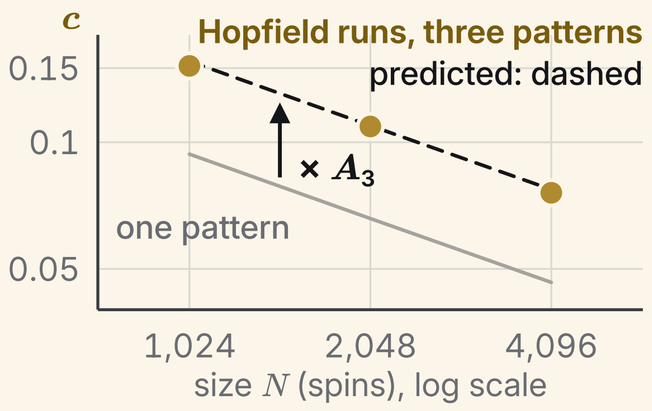


$$c_{\mathrm{pred}} = A_p\, c_{\mathrm{CW}}, \quad A_p = \frac{\mathbb{E}\max_\mu |Z_\mu|}{\mathbb{E}|Z|}$$

Bigger overlap, stronger bias.

Eq. 6 · assumes independent Gaussian amplitudes

5 p = 3: within 3% at three sizes.



1.662 × 0.0659 = 0.1095 predicted
measured 0.1090 **[0.1059, 0.1123]**

pooled over 50 pattern sets · N = 2,048 above · Table 1
size exponent 0.50 [0.47, 0.53]; exact CW 0.506

6 Four patterns: 6% low.

p	N	measured ÷ predicted	ratio
1	1,024		0.990
1	2,048		1.001
1	4,096		1.015
2	2,048		1.011
3	1,024		0.975
3	2,048		0.995
3	4,096		0.982
4	**2,048**		**0.940**

0.9 1.0 = predicted

Holds for pooled runs, not one network: single sets depart beyond sampling noise.

Fig. 2b · bars: 95% intervals · Sec. 5
protocol: factor not supported as derived

Kondepudi & Nelson 1983 · Kondepudi & Gao 1987 · Amit et al. 1985 · Berglund & Gentz 2002 · Levin et al. 2010